Kuku AI Generalist Assignment — Problem 1

Prototype: Kuku Promo Finder Enhanced | **Show:** I Married a Broken Soldier. He was a Duke

Problem and approach

This prototype helps editors find 20–60 second microdrama promos from timestamped screenplay text. It parses episode/scene/event information, proposes continuous clips and two-segment montages, ranks them, and exports exact cut points, hook/cliffhanger evidence, rationales and audit files. Script timecodes are assumed to align with the final edit.

Candidate generation and ranking

Candidate windows start at timestamped event boundaries and target 30/40/50/60 seconds. Transparent scoring considers dramatic hook language, ending tension, dialogue/action density and cut alignment. The montage stage joins two non-overlapping segments (same or different episodes) and enforces a combined duration of 20–60 seconds. It suppresses near-duplicate candidates.

Gemini model failover

When GEMINI_API_KEY is configured, the top 60 candidates are reviewed in batches of 10 for first-three-second hook, unresolved cliffhanger, standalone comprehension and escalation. Primary: **gemini-3.8-flash**, attempted immediately and retried twice after 2-second waits for transient quota/capacity/server/network errors. After three failures, fallback: **gemini-3.5-flash**, with the same three-attempt policy. Permanent errors (invalid key/request/model) are not retried. If both models fail, deterministic ranking remains available and errors are logged. Model IDs are checked against Google's current Gemini API model documentation.

Full-pack execution

50/50 episodes processed; **335** single-segment candidates; **30** montage candidates; **365** total. Seven automated tests pass, including mocked retry/fallback tests. The delivered full-pack raw outputs were generated in heuristic mode without a supplied private API key; a live Gemini call still needs verification with the user's key.

Rank	Candidate	Duration	Score
1	Episode 38	40.0s	5.747
2	Episode 46	60.0s	5.465
3	Multi-segment: Episode 21 +	60.0s	5.212
4	Episode 20	50.0s	5.191
5	Episode 28	60.0s	5.184

Scores are editorial triage scores, not calibrated probabilities.

Quality assurance, limitations and handoff

Known limitations

Text-only analysis cannot verify acting, facial expressions, shot composition, transitions, audio, music, visual continuity or whether a cut is understandable in the finished video. Heuristics can misread dramatic language; montage suggestions can combine unrelated moments; LLM judgments can be wrong. Human review of every selected video cut remains required.

How to validate

Have editors label top-3/top-10 candidates as usable or unusable and record reasons. Measure precision@3/10, recall against editor-selected promos, timestamp correction rate, montage continuity rejection rate and editor review time saved. Compare against manual selection on the same episodes.

Tools, deployment and disclosure

Python handles parsing/scoring/exports; Streamlit provides the UI; Google's google-genai SDK provides optional semantic reranking. The base pipeline runs without an API key. Gemini calls may be subject to account quota, billing and model availability. The app records model usage and failures. Keep the API key in environment variables or Streamlit Secrets, never in Git. Do not publish Kuku's confidential screenplay input pack.

Run

`py -3 -m venv .venv` → activate → `pip install -r requirements.txt` → `python -m unittest discover -s tests -v` → set `GEMINI_API_KEY` → `python promo_finder.py --input data --output outputs` → `streamlit run app.py`. See README for Windows PowerShell and Streamlit Cloud Secrets.

AI disclosure: ChatGPT assisted with implementation, tests and documentation. Gemini semantic review is optional and must be checked by a human editor.